

Weekly Diary – Week 21

Internship Organization: Vikaspedia

Duration: 26/05/2026 – 31/05/2026

Objective of the Week

The objective of this week was to enhance the indexing system by improving search functionality, optimizing multilingual content processing, and performing testing on indexed data from different Indian languages.

Date- 26/May

Worked on improving the indexing structure to support efficient search and retrieval operations. Studied tokenization techniques for Indian languages and analyzed how words are segmented differently across various scripts. Implemented improvements in text storage and retrieval mechanisms to support faster indexing operations.

Date- 27/May

Developed search functionality for retrieving indexed content based on user queries. Implemented keyword matching and experimented with multilingual search scenarios. Tested searches using Hindi and Punjabi content and analyzed how indexing quality affected retrieval accuracy. Made improvements to query processing for better search results.

Day- 28/May

Focused on optimizing multilingual text processing and reducing indexing overhead. Improved content normalization techniques to ensure consistency across indexed documents. Conducted performance testing using larger datasets and monitored indexing speed, storage utilization, and retrieval response times.

Date- 29/May

Performed extensive testing of the indexing and search modules using content from multiple Indian languages. Verified that search results were relevant and correctly mapped to indexed documents. Identified and resolved issues related to Unicode handling and special character processing during indexing.

Date- 30/May

Worked on improving the overall system reliability by adding validation and error-handling mechanisms. Implemented logging functionality for indexing operations and monitored processing statistics. Conducted end-to-end testing of the complete workflow including content extraction, language detection, indexing, and search

Date- 31/May

Prepared project documentation describing the indexing workflow, language processing pipeline, search architecture, and testing methodology. Reviewed all implemented modules and performed final testing to ensure stable operation across multiple Indian languages. Documented observations and future enhancement possibilities for multilingual search capabilities.

Conclusion

By the end of this week, the Indian Language Web Index system was capable of processing, indexing, and retrieving multilingual web content efficiently. Significant improvements were made in language detection, indexing accuracy, search functionality, and overall system performance, resulting in a stable foundation for large-scale multilingual information retrieval.